

RANSAC

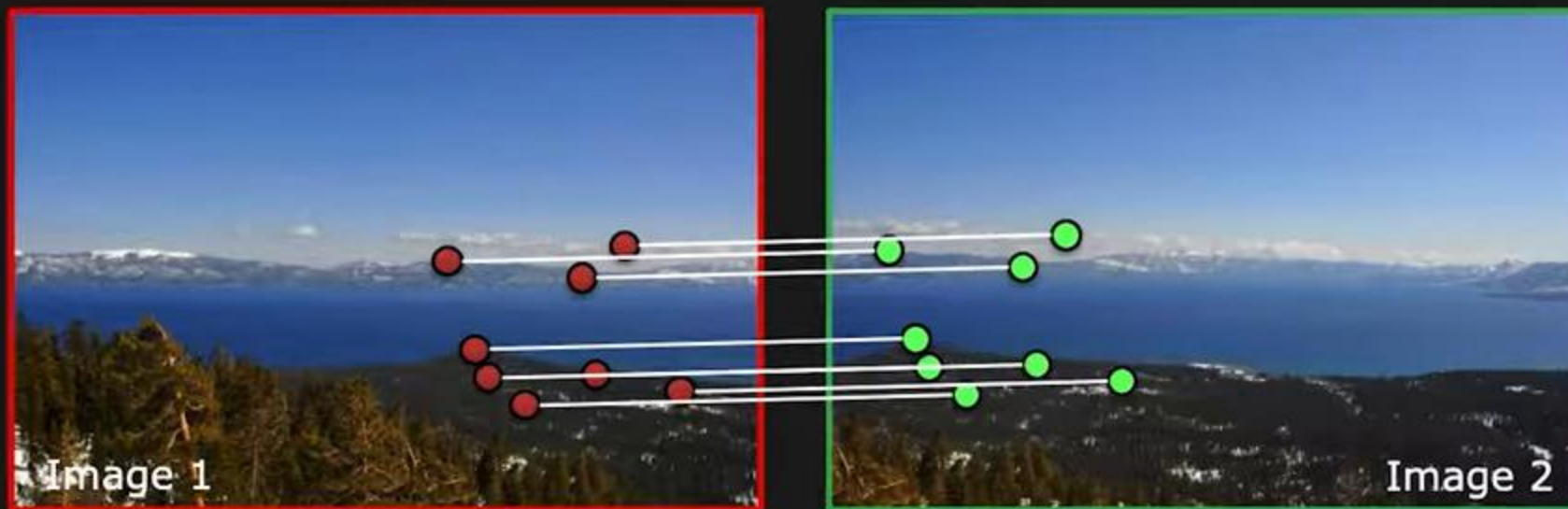
Dealing with Outliers: RANSAC

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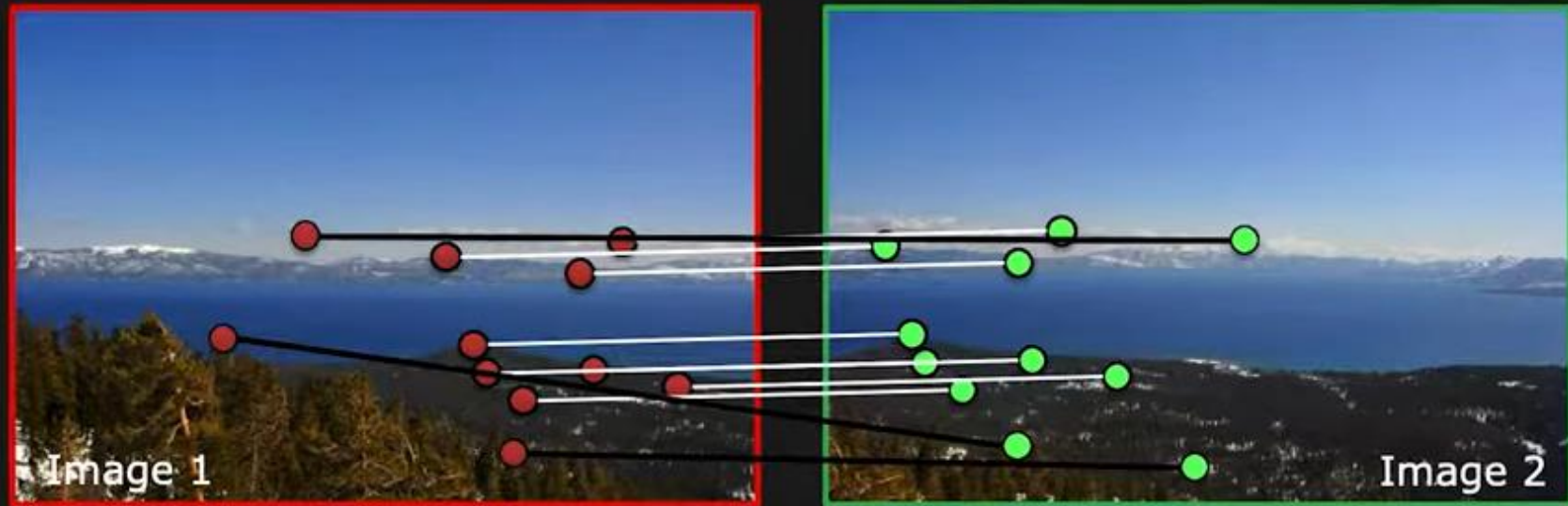
Topic: Image Stitching, Module: Features

First Principles of Computer Vision

What Could Go Wrong?



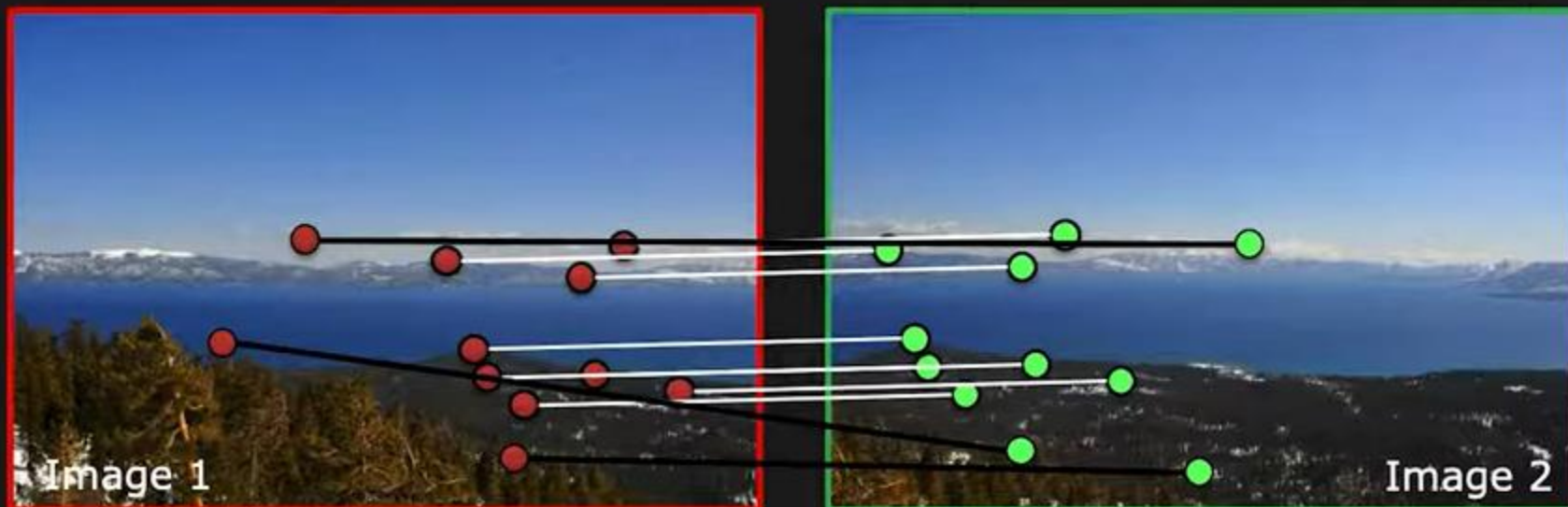
What Could Go Wrong?



Outliers!

We need to robustly compute transformation in the presence of wrong matches.

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Outliers!

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If number of outliers $< 50\%$, then RANSAC to the rescue!

RANdom SAmple Consensus

General RANSAC Algorithm:

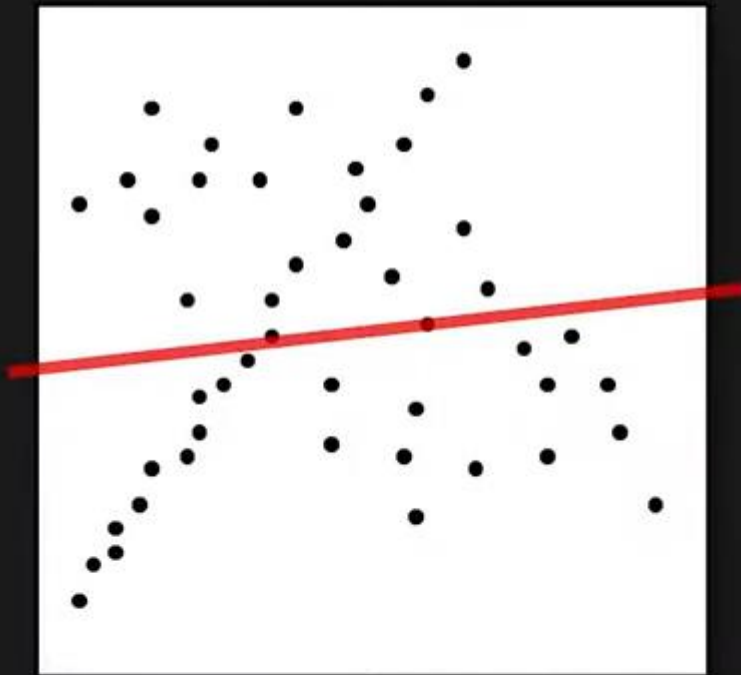
1. Randomly choose s samples. Typically s is the minimum samples to fit a model.
2. Fit the model to the randomly chosen samples.
3. Count the number M of data points (inliers) that fit the model within a measure of error ϵ .
4. Repeat Steps 1-3 N times
5. Choose the model that has the largest number M of inliers.

For homography:

$s = 4$ points. ϵ is acceptable alignment error in pixels.

RANSAC Example: Line Fitting

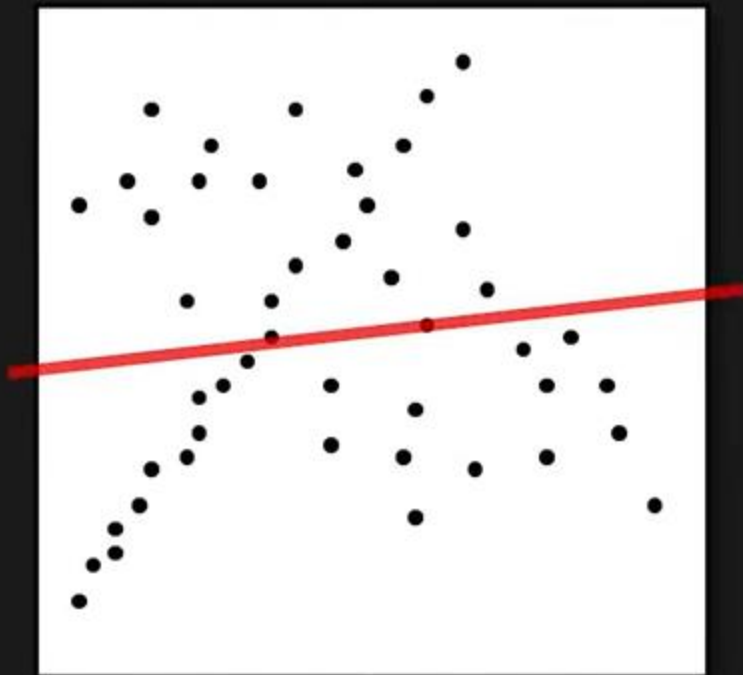
Robust line fitting:



Least Squares Fitting

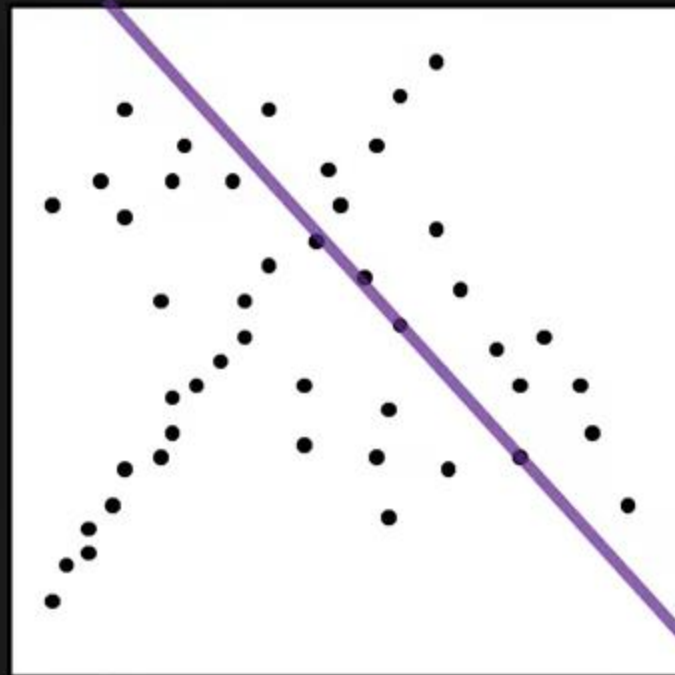
RANSAC Example: Line Fitting

Robust line fitting:



Least Squares Fitting

Inliers: 2

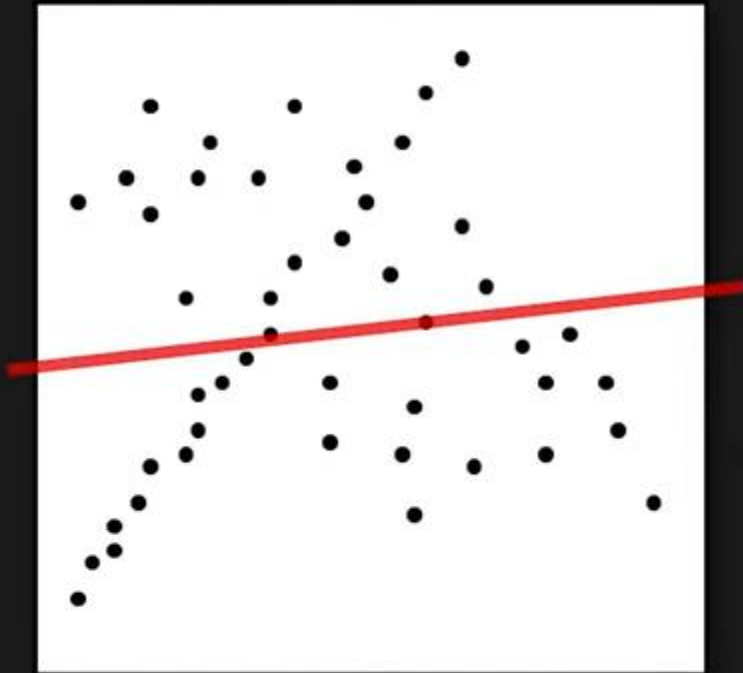


RANSAC Iteration 1

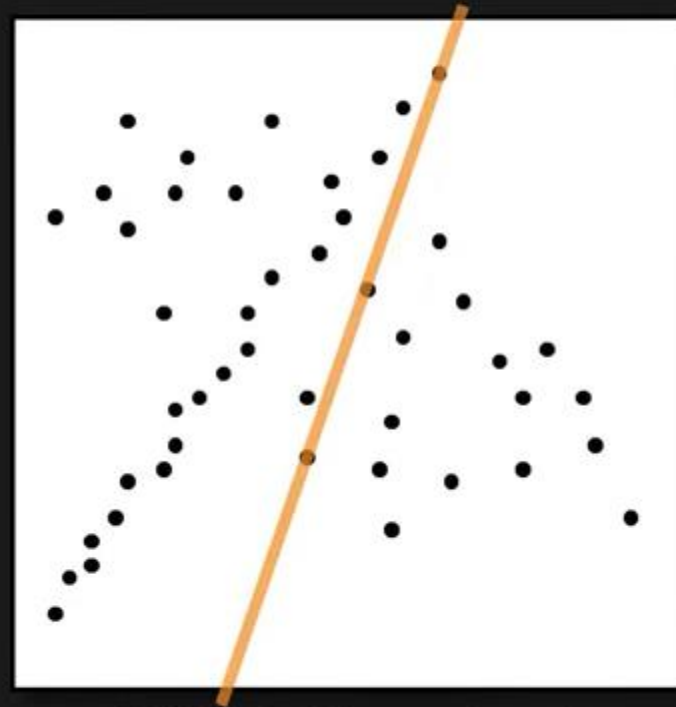
Inliers: 4

RANSAC Example: Line Fitting

Robust line fitting:



Least Squares Fitting

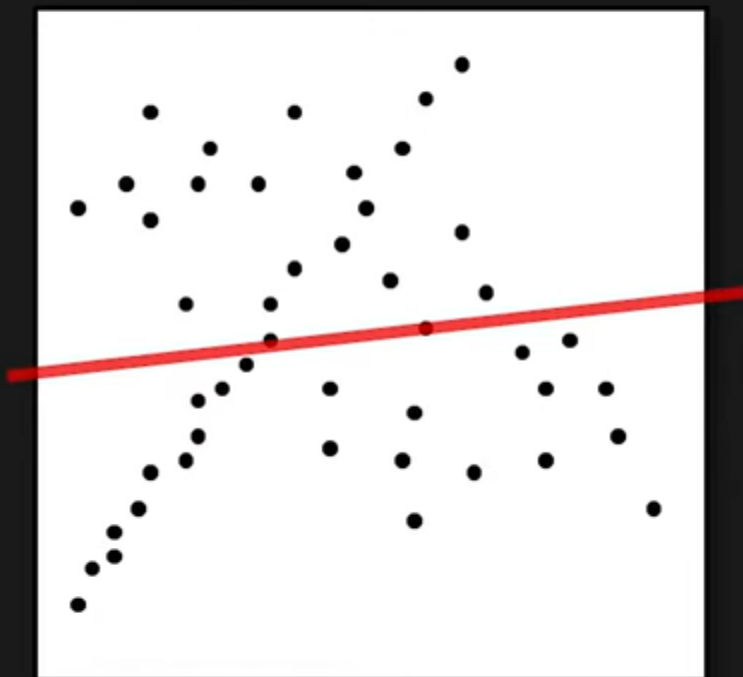


RANSAC Iteration 2

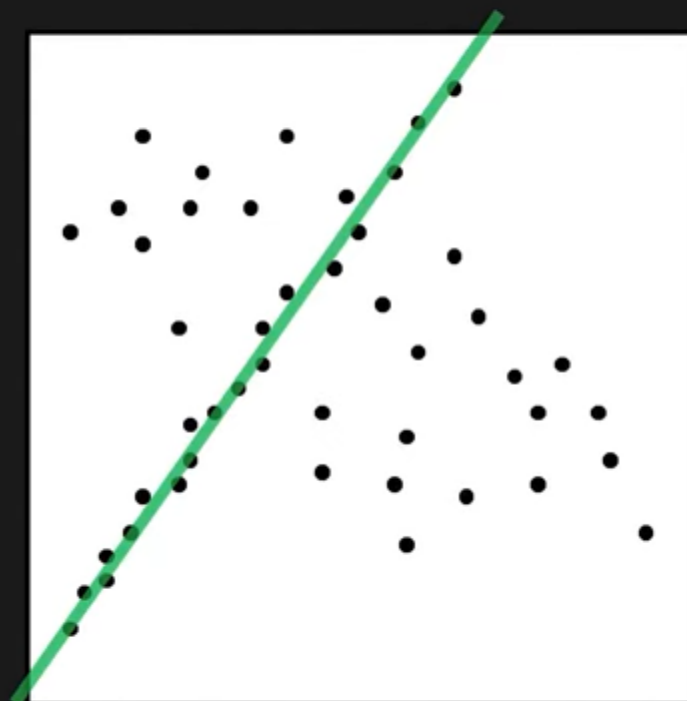
Inliers: 3²

RANSAC Example: Line Fitting

Robust line fitting:



Least Squares Fitting



RANSAC Iteration i

Inliers: 20

Reference

- <https://www.youtube.com/watch?v=EkYXjmiolBg>